北京邮电大学 本科毕业设计（论文）初期进度报告

Project Early-term Progress Report

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论文题目 Project Title	AI Assisted Epilepsy detection using EEG signals				

1. Literature Review and Critical Analysis

Epilepsy is a chronic non-communicable disorder of the brain that affects approximately 50 million people worldwide. The hallmark of epilepsy is recurrent seizures, which are brief episodes of involuntary movement. EEG remains the gold standard for diagnosis. However, manual inspection of long-term EEG recordings is tedious, time-consuming, and prone to human error. Consequently, Automated Seizure Detection Systems (ASDS) using Deep Learning have gained significant attention.

In preparing for this project, I have conducted a systematic review of recent literature, focusing on the evolution from traditional machine learning to advanced Deep Learning architectures. My analysis highlights a progression from feature-engineering-heavy approaches to end-to-end learning models.

1.1 Convolutional Neural Networks (CNN) in EEG Analysis

Convolutional Neural Networks have been extensively and successfully applied to EEG signal analysis. Unlike their application in static images, 1D-CNNs in EEG processing utilize convolutional filters to extract local temporal and spectral features effectively. One of the foundational works in this domain is by Acharya et al. (2017), who proposed a deep Convolutional Neural Network for the automated detection of seizure segments[1]. Their model utilized a 13-layer deep CNN architecture to automatically learn features from raw EEG signals, achieving high accuracy.

Recent Developments: Advanced architectures such as CodimNet[2] demonstrate the power of convolutional filtering in capturing complex non-linear features from raw EEG data without manual feature engineering. CodimNet illustrates how multi-scale convolutions can handle the high-dimensional nature of EEG signals robustly.

My Perspective: While CNN-based approaches like CodimNet are highly effective at feature extraction, they primarily focus on local patterns within the receptive fields. For this specific project, I aim to investigate Recurrent Neural Networks, specifically Bi-LSTM, to explore their capability in modeling long-term temporal dependencies across the entire seizure event. This provides a comparative perspective to the dominant CNN-based methods.

1.2 The Shift to Recurrent Neural Networks (RNN) and LSTM

Addressing the temporal limitations of CNNs, Hussein et al. (2018) introduced a robust Long Short-Term Memory (LSTM) network [3]. Their approach focused on analyzing EEG signals that contain artifacts (muscle activities, eye movements) and white noise. They employed a strategy of segmenting the dataset and using average pooling before the final classification to handle the non-stationary nature of the signals.

My Insight & Critique: The most valuable takeaway from Hussein's research is the importance of robustness against noise. In clinical settings, EEG data is rarely clean. However, standard LSTMs

process data in a single direction (past to future). In my view, for a retrospective analysis of EEG recordings (where the entire signal is available), limiting the model to uni-directional context is inefficient. We should exploit the context from both past and future data points to make a more informed classification for any given timestamp.

1.3 Bidirectional and Hybrid Architectures
Building on the concept of temporal context, Thara et al. (2019) proposed a Stacked Bidirectional LSTM (Bi-LSTM) model [4]. Their study on the Bonn dataset achieved 99% accuracy by stacking two layers of Bi-LSTMs, allowing the network to learn from the input sequence in both forward and backward directions.

My Insight & Critique: This paper is pivotal for my project. The success of Bi-LSTM confirms my hypothesis that bidirectional information flow is critical for high-fidelity seizure detection. However, a gap remains: deep models like stacked Bi-LSTMs are computationally expensive. Ullah et al. (2018) proposed a Pyramidal 1D-CNN to reduce parameter size (using only 39% of traditional parameters) while maintaining accuracy [5].

Synthesis for My Project: Balancing accuracy and computational efficiency is key. Therefore, my project will focus on optimizing a Bi-LSTM architecture specifically for the Bonn dataset. Unlike many studies that merge multiple datasets (e.g., CHB-MIT with Bonn) to increase data volume, I will strictly adhere to the project constraint of using the Bonn University Dataset alone. This forces the challenge of designing a model that generalizes well on a smaller, specific dataset without overfitting—a problem I intend to solve by implementing rigorous data augmentation and regularization techniques (like Dropout), as suggested in the literature.

2. Summary of Progress
I have successfully completed the initial phase of the project, moving from theoretical understanding to practical environment setup and data analysis.

2.1 Theoretical Foundation and Skill Acquisition

Deep Learning Architectures: I have completed an in-depth study of Recurrent Neural Networks and their variants. Specifically, I focused on the "Vanishing Gradient Problem" inherent in simple RNNs and how the Gating mechanisms in LSTMs (Input, Forget, and Output gates) and GRUs (Update and Reset gates) solve this.

EEG Signal Characteristics: I familiarized myself with the physiological basis of EEG, understanding the difference between *ictal* (seizure), *inter-ictal* (between seizures), and *normal* states, and how these manifest in signal amplitude and frequency.

2.2 Dataset Acquisition and Exploratory Data Analysis (EDA)
I have selected the **University of Bonn EEG Dataset** as the data source for this project.

- **Data Structure:** I successfully acquired the complete dataset, which consists of 5 subsets (Z, O, N, F, S), also referred to as Sets A, B, C, D, and E.
    - **Sets A & B:** Surface EEG from healthy volunteers (Eyes Open/Closed).
    - **Sets C & D:** Intracranial EEG from epilepsy patients during seizure-free intervals (Inter-ictal).
    - **Set E:** Intracranial EEG during seizure activity (Ictal).
- **Data Processing:** The original data was in single-channel ASCII format. I wrote a Python script using pandas and numpy to:
    1. Iterate through the 100 text files in each folder.
    2. Merge them into a structured dataset of size 500×4097 samples.
    3. Visualize the signals using matplotlib.
- **Key Observation:** My initial visualization confirms distinct differences in amplitude. Set E (Seizure) shows significantly higher amplitudes (often > 1000 μV) and chaotic

patterns compared to Sets A and B (< 100 μV). This validates that the dataset is suitable for training a discriminative Deep Learning model.

2.3 Technical Implementation

Environment Setup: I have configured a local development environment using Anaconda. The core stack includes Python 3.8, TensorFlow 2.x (Keras API) for model building, and Scikit-learn for data splitting and evaluation metrics.

Preprocessing Pipeline: I have begun implementing a segmentation function. Since each recording is 23.6 seconds (4097 data points), feeding the entire sequence into an LSTM might cause memory bottlenecks. I am currently experimenting with segmenting these into smaller windows (e.g., 2048 or 1024 data points) to increase the number of training samples and improve the model's ability to detect short-duration anomalies.

3. Strategy for Addressing the Small Dataset Issue

A critical challenge in this project is the limited size of the Bonn University Dataset (500 trials total). Training deep learning models on such small data can lead to severe overfitting. To address this, I have formulated a comprehensive **Data Augmentation and Regularization Strategy**:

3.1 Data Segmentation with Overlap (Sliding Window)

Instead of feeding the entire 23.6-second recording (4097 data points) as a single sample, I will implement a **Sliding Window Segmentation** technique.

- **Implementation:** The continuous EEG signal will be sliced into smaller windows (e.g., 1 to 2 seconds).
- **Overlap:** Crucially, I will apply a **50% overlap** between consecutive windows.
- **Impact:** This technique will increase the dataset size from 500 samples to approximately **10,000+ training segments**. This massive increase in effective sample size is essential for the model to learn generalized features and converge properly.

3.2 K-Fold Cross-Validation

To ensure the reported accuracy is reliable and not a result of a lucky train-test split, I will employ **10-Fold Cross-Validation**. The dataset will be divided into 10 subsets; the model will be trained on 9 and tested on 1, repeating the process 10 times. The final performance will be the average of these 10 runs, providing a statistically robust metric.

3.3 Regularization Techniques

- **Dropout:** I will insert Dropout layers (rate = 0.3 to 0.5) within the Bi-LSTM architecture to randomly drop neurons during training, forcing the network to learn redundant representations and preventing reliance on specific noise patterns.
- **Early Stopping:** Training will monitor the validation loss and stop automatically when performance plateaus, preventing the model from memorizing the training data.

4. References

[1] U. R. Acharya, S. L. Oh, Y. Hagiwara, J. H. Tan, and H. Adeli, "Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals," *Computers in Biology and Medicine*, vol. 100, pp. 270–278, Sep. 2017.

[2] Z. Zhang, Y. Wei, X. Rao, *et al.*, "Electroencephalographic biomarker-guided early detection of Alzheimer's disease via cortically subdivided neurodynamic PINN," *Expert Syst. Appl.*, vol. 297, p. 129353, 2026.

[3] R. Hussein, H. Palangi, R. K. Ward, and Z. J. Wang, "Optimized deep neural network architecture for robust detection of epileptic seizures using EEG signals," *Clinical Neurophysiology*, vol. 130, no. 1, pp. 25–37, 2019.

[4] D. K. Thara, B. G. PremaSudha, and F. Xiong, "Epileptic seizure detection and prediction using stacked bidirectional long short term memory," *Pattern Recognition Letters*, vol. 128, pp. 529–535, 2019.

[5] I. Ullah, M. Hussain, E. Qazi, and H. Aboalsamh, "An automated system for epilepsy detection using EEG brain signals based on deep learning approach," *Expert Systems with Applications*, vol. 107, pp. 61–71, 2018.
[6] S. Roy, I. Kiral-Kornek, and S. Harrer, "ChronoNet: A Deep Recurrent Neural Network for Abnormal EEG Identification," in *Proc. IEEE Winter Conf. on Applications of Computer Vision (WACV)*, 2018, pp. 1–10.
[7] M. Golmohammadi et al., "Gated Recurrent Networks for Seizure Detection," *Neural Engineering Data Consortium*, Temple University, Philadelphia, PA, USA, 2019.

是否符合进度? On schedule as per GANTT chart?

YES. I have completed the literature review, selected the methodology (Bi-LSTM), acquired and preprocessed the Bonn dataset, and set up the coding environment. This aligns perfectly with the initial phase of the project timeline.

下一步 Next steps:

1.Construct and compile the Bi-Directional LSTM model using TensorFlow/Keras.
2.Perform initial training on the 2-class problem (Normal vs. Seizure) and record baseline metrics.
3.Implement data segmentation techniques to handle the 4097-point sequences effectively.